

Obligatory Control into Propositional Complements

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Abstract: This paper considers three puzzles about obligatory control (OC) with so-called propositional verbs like *claim* in English (*Chris claimed (to the judge) to be innocent*). The first is why control is possible with *claim*, but not with other propositional verbs like *believe* and *say*. The second is why in French control is possible with propositional verbs more broadly—with analogs of *believe* and *say* as well as *claim*. The third is why propositional verbs are strictly subject control predicates—even when an object is also present. These facts can be explained by a theory in which OC is induced by thematic uniqueness: OC happens when the matrix clause expresses that one of its arguments bears a certain thematic role to a certain event, and the embedded clause expresses that its PRO subject bears the same thematic role to what turns out to be the same event. With desiderative verbs like *decide*, it is the agent role that has this effect, but with propositional verbs it is the *about* role. This analysis also accounts for other little-recognized properties of control with propositional verbs: it is disrupted by an overt *about* phrase, it never shifts when the embedded clause is passive, and it is incompatible with the matrix verb being an impersonal passive. The result is a general theory of how OC with propositional verbs is and is not like OC with desiderative verbs.

Key words: obligatory control, PRO, infinitival clauses, propositional verbs, English, French, thematic uniqueness

1. Introduction: Three Puzzles for a Theory of Control

Obligatory control (OC) is a classic topic in generative linguistics, tracing back to Rosenbaum (1967), and many linguists have studied it over decades. Nevertheless, there are things that we do not know about it—even in well-studied languages like English and French. This is particularly true when one gets beyond the most canonical verbs that take infinitival complements, such as ‘promise’, ‘persuade’, and ‘decide’—verbs in the so-called desiderative class according to the typology of (possible) control verbs that is put forward in Idan Landau’s invaluable overviews of this topic (e.g., Landau 2013: 89, Landau 2015: 7).

This investigation shifts the focus somewhat, to put some new attention on the so-called *propositional* class of verbs: verbs like ‘claim’, ‘think’, and ‘say’ (Landau 2015: 7). These verbs

can be characterized as ones whose clausal complement expresses a proposition: something that has the property of being true or false. *Claim* in English is in this class, whereas *decide* is not, as shown by the fact that (1a) is felicitous, whereas (1b) is odd.¹

- (1) a. John_i claimed [PRO_i to have paid the fine], which was false. (He didn't actually pay it.)
- b. #Mary_i decided [PRO_i to pay the fine], which was false. (She won't in fact pay it.)

There are three notable puzzles about OC with propositional verbs that warrant a fresh look. The first is a language-internal puzzle that arises in English. This is the often-noted but almost never explained fact that *claim* is (almost) the only propositional verb that allows an infinitival complement with OC. Other canonical verbs from this class do not allow this. So (1a) in English is fine, but (2a,b) are not.²

- (2) a. *Tom said to have paid the fine. (cf. Tom said that he paid the fine.)
- b. *Peter thinks to have paid the fine. (cf. Peter thinks that he paid the fine.)

What is it about *claim*, then, that makes it stand out in its class in this way?

The second puzzle about control with propositional verbs is a crosslinguistic one. Other languages rather closely related to English do allow OC with a much wider range of propositional verbs. This includes analogs of *say* and *think* as well as *claim* (if any), and also lower frequency propositional verbs like *affirm*. Thus, (3a-d) are all possible in French.

- (3) a. J'ai réclamé avoir demandé l'annulation. (Google search, 1/20/2025)³

¹ For English sentences, the gender of each person-denoting noun phrase was determined randomly, by flipping a coin, except where I deemed it helpful to have DPs of two different genders, to help disambiguate pronouns.

² I will not be concerned with the exact boundaries of the class of propositional verbs, however. I will consider any verb that seem similar to 'claim', 'think' and 'say' along some semantic dimension, i.e. verbs of cognition and communication. I show below that there are more propositional verbs that allow OC in English than often thought, once one includes verbs that select gerundival complements rather than infinitival ones.

³ In fact, French does not have a close analog of 'claim'. I use *réclamer* in (3) for illustrative purposes, since it is cognate with and can be translated as 'claim'. However, the verb seems to be relatively rare, at least in the OC construction, and I found only a few examples with Google. The examples in (3) have full glosses and translations in line with the Leipzig glossing conventions. However, in later sections I collapse these into one line, taking advantage of the fact that French word order is very similar to English (except for the position of pronominal objects) and many of the key words—especially the matrix verbs—are cognate.

I-have claimed have.INF requested the-cancellation

‘I claimed that I had requested the cancellation.’

- b. Je crois/affirme avoir fait une erreur. (Kayne 1984: 112)

I believe/affirm have.INF made a mistake

‘I believe/affirm that I have made a mistake.’

- c. Jean dit l’avoir vue à la fête. (Reed 2014: 264 n.8 (ii))

Jean says her-have.INF seen.F at the party

‘Jean says that he saw her at the party.’

Like French in this respect are Italian (Rizzi 1982: 80) and German (Wurmbrand 2014: 408n4) among others. Wurmbrand says that “Crosslinguistically, propositional control is very common.” Hebrew, in contrast, is more like English in disallowing OC with propositional verbs (except for the noncanonical propositional verb ‘pretend’) (Idan Landau p.c. 2024). English is thus not a unique outlier either. What then is it about French that *croire* and *dire* allow OC whereas *think* and *say* in English do not?

Although this difference is a classical observation going back to the GB period, classical accounts for it were quite limited. Chomsky (1980: exx (80)-(81)) noted this contrast between English and French/German in that first study to use notions of government and case assignment; see also Chomsky (1981), Kayne (1984: ch. 5) and Rizzi (1982: ch. 3), among others. A leading idea at the time was that *believe* in English selects a reduced complement (S, now TP) whereas *croire* in French selects a full clausal complement (S’, now CP). Nonfinite clauses have PRO subjects, PRO could not be governed, and CP but not TP prevented the matrix verb from governing PRO. As a result, control was possible in (3) but not in (2b). But this cluster of ideas involving government and its effect on PRO is obsolete within the Minimalist Program.

Moreover, there was always a stipulative element to the account, simply saying that one verb selects one type of clause and another a different type (as recognized by Kayne (1984: 113)).

However, more contemporary versions of control theory do not have much if anything more to say about this. One of the most influential ideas has been that OC depends in some way on the tense-aspect value of the infinitival clause in relation to the matrix clause, such as the distinction between future/tensed infinitives and nonfuture/nontensed ones: see Stowell (1982), Chomsky & Lasnik (1993), Bošković (1997), Martin (2001), Landau (2001, 2004), Wurmbrand (2003) for different versions of this idea. However, Wurmbrand (2014) demonstrates with considerable care that the infinitival complements of the control verb *claim* and the noncontrol (but rather exceptional case marking (ECM)) verb *believe* have exactly the same tense-aspect properties. There is no difference between them along these lines that can explain why *claim* is a control verb and *believe* is not. As a result, Wurmbrand (2014: 410) conjectures that this must be a direct lexical/selectional difference—similar to the GB theory. Nor has the movement theory of control in the tradition of Hornstein (1999) had anything to say about these contrasts within the set of propositional verbs, as far as I know. Finally, semantically-based theories of control like that of Jackendoff & Cullicover (2003), which are designed to account for OC with verbs like *decide* and *promise* that take “actional” complements, often stumble on the propositional control verbs, whose complements are not semantically restricted in this way (see Landau 2013: 135).

The third puzzle about propositional verbs that calls for attention is the little-realized fact they systematically license subject control even in the presence of an object (direct or indirect). In English, this can be seen with *claim*: when a goal argument is included, the subject can still control PRO in the CP complement ((4a)), whereas the goal argument cannot control ((4b)).

- (4) a. David_i claimed to the judge [PRO_i to have paid the fine].

- b. *Pam claimed to the judge_i [PRO_i to have treated her unfairly].

This goes against the received wisdom that *promise* in English is very special as virtually the only ditransitive subject control verb, or (more accurately) as the most common of a small class of verbs of “commitment” that include synonyms of *promise* (Sag and Pollard 1991, Jackendoff and Culicover 2003). For English, this might not seem like too big a gap, since considering propositional verbs as well as desiderative verbs may only increase the stock of subject control verbs by one (but see below). But it is a bigger deal when we include French. Since French has many propositional verbs that allow control, it also has many propositional verbs that allow subject control even in the presence of an object. (5) gives sampling. Note that this includes verbs that take a direct object plus infinitive (‘inform’, marginally ‘convince’) as well as verbs that take an indirect object plus infinitive (‘say’, ‘affirm’).

- (5) a. (Marie,) le portier l’a informé avoir trouvé son sac. (4 consultants)

(Marie) the doorman her.ACC-has informed have.INF found her bag

‘As for Marie, the doorman informed her that he found her bag.’

- b. (Marie,) le portier lui a dit avoir trouvé son sac. (4 consultants)

(Marie) the doorman her.DAT has told have.INF found her bag

‘As for Marie, the doorman told her that he found her bag.’

- c. ?Je l’ai convaincu d’être le bon candidate pour ce poste. (Google search⁴)

I him.ACC-have convinced of-be.INF the good candidate for this position

‘I convinced him that I was the best candidate for this position.’

- d. J’ai affirmé au juge avoir fait une grave erreur. (Viviane Deprez, p.c.)

⁴ This example was found by a Google search, and its meaning was clear in context, but my consultants did not like it much. (At issue here may be the degree to which ‘convince’ in French means ‘cause to believe’ as well as its meaning of ‘cause to intend’, whereas both meanings are perfectly possible in English.)

I-have affirmed to.the judge have.INF made a serious error

‘I affirmed to the judge that I made a serious error.’

I estimate that there are at least two dozen subject control verbs in French in this semantic domain, with various degrees of productivity. The question arises, then, why propositional verbs generally induce subject control, even in the presence of a possible object controller, whereas desiderative verbs more often have object control. This is a new problem for theories of OC based on the Minimal Distance Principle, including the movement theory of control.

This paper addresses these three puzzles concerning OC with propositional verbs. It does so by developing and applying a recent proposal about the nature of obligatory control by Baker (2024), which flows out of his study of serial verb constructions and constructions with logophoric operators. Unlike most previous accounts, this one is based on the matching of thematic roles between the controller in the matrix clause and the controllee in the embedded clause. For example, (6a) is an ordinary example of subject control with *promise*, where the agent-subject of the matrix clause controls the agent-subject of the embedded clause. In contrast, (6b) is an example of so-called *control shift* (Rosenbaum 1967: 92 n.13, Farkas 1988, Sag and Pollard 1991, Panther and Köpcke 1993, Landau 2013: Sec. 5.1.2), in which the subject of the embedded predicate is a goal/benefactee argument, not an agent. It can be interpreted as being controlled by the goal/benefactee argument of the matrix verb *promise*—an instance of object control rather than subject control.

- (6) a. Mary_i promised John [PRO_i to finally give him his due at the shareholders’ meeting].
b. Mary promised John_i [PRO_i to finally be given his due at the shareholders’ meeting].

This theory is reviewed in section 2. It follows from Baker’s proposal that there needs to be a certain kind of overlap in meaning between what is expressed in the matrix clause and what is

expressed in the complement clause for OC to take place. Section 3 discusses how this can give us a new paradigm for explaining the badness of control with many propositional verbs, as in (2): if the right kind of overlap in meaning is not present, then control does not happen, even if the matrix verb selects for an infinitival clause with the right kind of complementizer and favorable tense-aspect properties. Section 4 takes up the question of why *claim* is different in English. I show that *claim* has a different kind of meaning overlap with its complement: the matrix verb has a covert *res* argument (a null *about* phrase) that links up with the same meaning in the complement clause. This can be seen as a generalization of Baker's account to a new kind of thematic role. Section 5 then turns to French, and why it is different from English in this respect. I argue against the initially plausible hypothesis that the difference lives in the nature of the infinitival clause. Instead, I claim that French propositional verbs systematically allow a *res* argument to be included in the grammatical representation in a way that English verbs other than *claim* (and a few others) do not. Indeed, French verbs like 'believe' and 'say' with OC behave like *claim* in English in a cluster of ways. Section 6 presents my conclusions.

2. Obligatory Control as Induced by Thematic Uniqueness

Baker (2024) motivates the idea that thematic role matching is an important driver of OC using serial verb constructions (SVCs) in some West African languages such as Ewe (Collins 1997) and Edo (Stewart 2001). SVCs are clauses that have more than one thematic verb. In an important class of SVCs, the second of two verbs in the construction has a null theme argument which must be understood as being coreferential with an argument of the first verb. Collins (1997) argues that this theme argument is syntactically represented by a *pro*, and that it enters into a relationship of obligatory control. A crucial question, then, is which argument of the first

verb (V1) is understood as the controller of the null theme argument of the second verb (V2)?

Empirically, the answer is that the controller must be the theme argument of the first verb—not its agent argument, or its goal/source argument, even in cases where having the agent or the goal/source argument as the controller would make sense semantically. This is seen in (7).

(7) a. Wo ɖa fufu ɖu. (Ewe, Collins 1997)

they cook fufu eat

‘They cooked fufu and ate it.’

b. Ekpe a fo kɔpo yi xɔ-me. (Ewe, Collins 1997)

rock FUT hit cup go room-in

‘A rock will hit the cup into the room.’ (*Not*: ‘A rock will hit the cup and then go into the room.’)

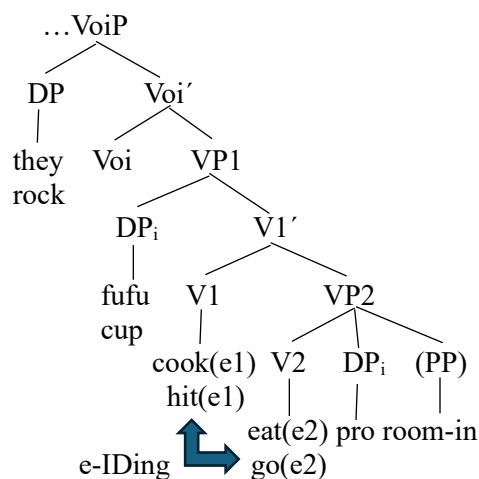
c. Uyi vbɔ ɔkhɔkhɔ igan khiɛn. (Edo, Stewart 2021: 127)

Uyi pluck chicken feather sell

‘Uyi plucked feathers from the chicken and sold them/*it.’

Collin’s (1997) structure for this kind of SVC is given in (8) (slightly updated). This shows the second verb heading a VP projection that functions as the complement of the first verb. This VP contains a *pro* theme argument, which is controlled by the theme argument of the first verb, generated in Spec of the larger VP. (The first V moves to Voice, giving the correct word order.)

(8)



The appealing generalization here is that the controller of the complement of the second verb must have the same thematic role (theme) as the controllee. Baker (2024) then asks what known principle of grammar can deliver this generalization in a natural way. The answer, he suggests, is the principle thematic uniqueness from Carlson (1984)—his formulation of part of Chomsky’s (1981) Theta Criterion. This principle can be stated as in (9); for additional empirical and theoretical discussion, see Dowty (1989, 1991), Parsons (1995), and recently Paillé (2024).⁵

- (9) *Thematic uniqueness*: If x bears thematic role θ_a to event e_1 , and y bears the same thematic role θ_a to event e_2 and $e_1=e_2$, then $x=y$.

This principle is relevant to an SVC structure like (8) because V1 and VP2 are both predicates of events. As such, their combination can be interpreted conjunctively, by event identification in the sense of Kratzer (1996). Evidence that event identification takes place in SVCs is that it is impossible to have an adverb in this type of SVC that modifies the pushing event but not the falling event or vice versa, as shown in (10). Linguistically, there is one event that is a pushing and a falling, and that one event is quick.⁶

- (10) Ozó (gié!gié) sú!á ògò (*gié!gié) dé. (Edo, Stewart 2001: 24)

Ozo quickly push bottle (*quickly) fall

‘Ozo (quickly) pushed the bottle down.’ (only: the event of pushing+falling was quick)

Given this additional assumption, the semantic interpretation of (10) can proceed roughly as in (11).

Crucially, there is an event of pushing that is also an event of falling (or closely related to it; see note 5),

⁵ Of the various explicit formulations of this principle in the literature, (9) is most like the one stated in Dowty (1989: 85 (22))—which he calls “Uniqueness of role-bearer”.

⁶ In this discussion I (following Baker 2024) gloss over the possibility that different subclasses of SVCs may have slightly different structures and interpretations (see Stewart 2001 for discussion). A more nuanced view might well also hold that falling and the pushing are not literally the same event but are both parts of a larger event, which must have a single well-defined theme (see Baker 2024: xx for some discussion). The crucial point for current purposes is not details of the mereology of events, but simply how SVCs point to the possibility of (9) playing an important role in OC relationships.

by event identification. The theme of the pushing is the bottle (from (11d)). And the theme of the falling is *pro* (from (11a)). Therefore, the referent of *pro* must equal the referent of ‘the bottle’ by (9). This induces obligatory control of *pro*, the argument of V2, by the theme argument of V1 in (7a-c) and (10).

- (11) a. $[[VP2]] = \lambda e \text{ fall}(e) \ \& \ \text{theme}(e) = \text{pro}$ (function application)
 b. $[[V1]] = \lambda x \lambda e \text{ push}(e) \ \& \ \text{theme}(e) = x$ (lexical meaning; Kratzer 1996)
 c. $[[V1']] = \lambda x \lambda e \text{ push}(e) \ \& \ \text{theme}(e) = x \ \& \ \text{fall}(e) \ \& \ \text{theme}(e) = \text{pro}$ (event identification)
 d. $[[VP1]]: \lambda e \text{ push}(e) \ \& \ \text{theme}(e) = \text{the.bottle} \ \& \ \text{fall}(e) \ \& \ \text{theme}(e) = \text{pro}$ (function application)

Therefore, $\text{the.bottle} = \text{pro}$, by (9)

The next question is whether this idea about control in SVCs has anything to offer to a theory of control of PRO in infinitival complements in languages like English, as for example with canonical examples of triadic verbs like in (12). Here (12a) has subject control, whereas (12b) has object control, although the structures of the two examples are superficially the same.

- (12) a. $\text{Mary}_i \text{ promised John}_k [\text{PRO}_i \text{ to come early}]$.
 b. $\text{Mary}_i \text{ ordered John}_k [\text{PRO}_k \text{ to come early}]$.

At first glance, an SVC-like analysis of controller choice does not look at all promising for these examples—especially (12b). First, the ordering is patently not the same event as the coming early. Rather, the two events can happen at different times, in different locations, in different possible worlds. One can be quick while the other is slow, and so on. Second, the thematic roles of the controller and the controllee do not match in this case: John is the goal argument of the ordering event, whereas PRO is the agent argument of the coming-early event. So (11) seems like a poor model for analyzing (12).

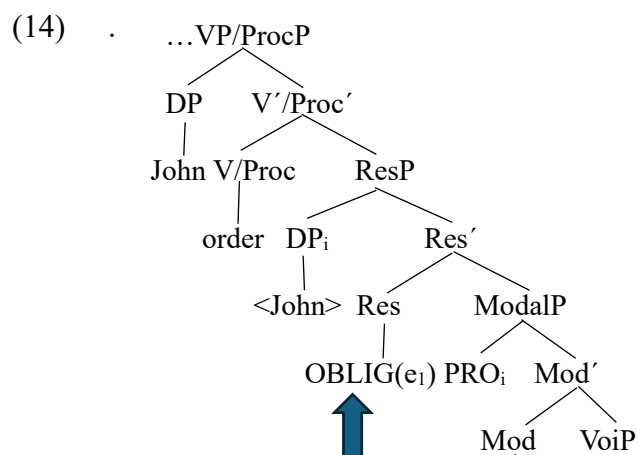
However, Baker (2024) bridges this gap by arguing that there is additional structure in both the embedded infinitival clause and in the matrix clause which is very relevant to understanding the OC. First, he follows Wurmbrand (2014) in claiming that there is a covert

modal head inside the infinitival complement. In the case of (12), it is a covert root modal with deontic force—something like *must*. This is motivated by the fact that not only is the coming event in an unrealized future with respect to the ordering event (Wurmbrand’s main concern), but the predicate in the complement of *promise/order* is subject to the same semantic restrictions as are the prejacent of root modal *must* or imperative morphology (see Jackendoff & Culicover 2003: 524-525, among others).

- (13) a. Catherine must be quiet/polite/#tall/#blonde. Be quiet/polite/#tall/#blonde!
 b. Sara promised/ordered Catherine to be quiet/polite/#tall/#blonde.

Given this, Baker assumes that the infinitival complement of *order* denotes an eventuality of its PRO subject being under an obligation to do something, whose content is expressed by its VP.

There is also more structure in the matrix clause, Baker claims. A verb like *order* expresses not only a certain process, but also a resulting state which is caused by that process. Following Ramchand (2008), Baker assumes that this resulting state is syntactically represented in the form of a ResP, the complement of a Proc (V) head. The verb *order* is then the result of these two pieces coming together, e.g. by head movement prior to vocabulary insertion. This assumption captures the fact that *Clyde ordered James to do something* implies that James must do that thing. On these assumptions, the structure of (12b) is (14), with a ModalP serving as the complement (not just a VP/TP) and a ResP in the matrix, the manifestation of a modest degree of predicate decomposition. ((14) gives just the VP of the matrix clause for simplicity. This VP would be the complement of a Voice head, with *Mary* as its specifier in the usual way.)



While it is true that the event of ordering is not the same as the event of coming early, we can say that the state of being obligated that is the result of the ordering is the same as the state of being obligated that is expressed by the embedded clause, including its covert modal. That is, in fact, what a sentence like (12b) means. Once these two states of being obligated are identified, then it follows from thematic uniqueness that the holders of those states are identified. The holder of the obligation according to the matrix clause is John, the goal of the ordering, by virtue of the lexical semantics of *order*. The holder of the obligation according to the embedded clause is PRO, by virtue of it being the specifier of the modal head (cf. Hacquard 2006, Hacquard 2010). Therefore, John is equated with PRO, resulting in object control in this case. The semantic composition is sketched in (15) following Baker (2024), where additional details can be found.

- (15) a. $[[\text{ModP}]] = \lambda e \exists z_c \text{holder}(e) = \text{PRO} \ \& \ \text{obligated}(e) \ \& \ \text{theme}(e) = z_c \ \& \ \text{content}(z_c) = \text{PRO come early}.$
- b. $[[\text{order-res}]] = \lambda y \lambda e \text{order-res}(e) \ \& \ \text{holder}(e) = y$ (lexical meaning of Res head)
- c. $[[\text{Res}']] = \lambda y \lambda e \exists z_c \text{order-res}(e) \ \& \ \text{holder}(e) = y \ \& \ \text{obligated}(e) \ \& \ \text{holder}(e) = \text{PRO} \ \& \ \text{theme}(e) = z_c \ \& \ \text{content}(z_c) = \text{PRO come early}.$ (event Iding)
- d. $[[\text{order-proc}]] = \lambda R \lambda x \lambda e \text{order}(e) \ \& \ \text{goal}(e) = \underline{x} \ \& \ \exists e_1 [e \rightarrow e_1 \ \& \ R(x, e_1)]$
(meaning of Proc head)

e. $[[\text{VoiceP}]] = \lambda e \text{ agent}(e) = \text{Mary} \ \& \ \text{order}(e) \ \& \ \text{goal}(e) = \text{John} \ \& \ \exists e_1 \ \exists z_c \ [e \rightarrow e_1 \ \& \ \text{order-res}(e_1) \ \& \ \textbf{holder}(e_1) = \textbf{John} \ \& \ \text{obligated}(e_1) \ \& \ \textbf{holder}(e_1) = \textbf{PRO} \ \& \ \text{theme}(e_1) = z_c \ \& \ \text{content}(z_c) = \text{PRO come early} \ (\text{Function Application} + \text{VoiceP})$
 $\text{PRO} = \text{John by Thematic Uniqueness}$

A goal of setting up the theory in this way is to derive the fact that *promise* participates in subject control in examples like (12a) from the same principles that give object control with *order* in (12b). The infinitival complement of *promise* is also a ModP headed by a covert deontic modal, as shown by the fact that the complement of *promise* has the same semantic restrictions on its predicate, as shown in (13). The core thematic roles in the matrix clause built around *promise* are also the same as with *order*: the subject is an agent and the object is a goal. The crucial difference is in the resulting state: whereas the result of an ordering event is that the goal of the ordering is under an obligation, the result of a promising event is that the agent of the promising is under an obligation. This fact about English lexical semantics can be observed apart from control using examples in which *promise* and *order* take a DP theme argument (or a finite CP), such as (16).

(16) a. Mary promised John a favor $\rightarrow$ Mary must do a favor.

b. ?Mary ordered John an arduous task $\rightarrow$ John must do an arduous task.

This can be expressed in the semantics by saying that the process head associated with *promise* relates to its ResP complement in a slightly different way from how the process head associated with *order* relates to its ResP. Its lexical semantics is (17); contrast this with (15d) above.

(17) $[[\text{promise-proc}]] = \lambda R \lambda y \lambda x \lambda e \text{ promise}(e) \ \& \ \text{goal}(e) = \underline{y} \ \& \ \exists e_1 \ [e \rightarrow e_1 \ \& \ R(\underline{x}, e_1)]$

Here the variable that gets plugged into the relation that comes from the ResP complement of Proc is *x*, distinct from the variable associated with the goal of the promising, whereas with

order these are instances of the same variable. This means that the argument of the resulting state is not identified with the goal of the promising; rather it will be identified with the agent of the promising, as part of the event identification that happens at the VoiceP level. In both cases, the holder of the resulting state is identified with the subject of the ModP complement, but how that holder relates to the arguments of the process head differs.

In this analysis, there is a control-like relationship which is stipulated to be different between *order* and *promise*, but that relationship is internal to the lexical semantics of *order* and *promise* (where it should be), rather than at the joint between the matrix clause and the embedded clause, where standard control of PRO happens. In essence, Baker (2024) is a relatively restrictive and syntacticized way of capturing an intuition expressed by Jackendoff & Cullicover (2003). They argue that what subject control with *promise* and object control with *order* have in common is that in both cases it is the one put under an obligation as a result of the matrix event that controls the understood subject of the embedded clause. Baker (2024) also relates this to argument sharing in SVCs (and to the control of logophoric operators). The account can be generalized to other kinds of control predicates by saying that ResP and ModP can express a variety of modal flavors, not just deontic MUST, but also ABLE (with *enable*), BE ALLOWED (with *permit*), INTEND (with *persuade*), and perhaps others.

One further concern of Baker (2024) is to account for the phenomenon of control shift. This is the much-discussed fact that, when the embedded clause is passive (or other nonagentive predicate), sentences with *promise* can shift to object control and sentences with *persuade* or *ask* (but not *order*) can shift to subject control, as shown in (18a,b). In contrast, *advise*, although superficially similar to *persuade*, resists shifting to subject control, and remains an object control predicate, as in (18c) (Panther and Köpcke 1993).

- (18) a. Mary_i promised John_k [PRO_k to be promoted to project manager by the new year].
 b. Mary_i persuaded John_k [PRO_i to be promoted to project manager by the new year].
 c. Mary_i advised John_k [PRO_{k,*i} to be promoted to project manager by the new year].

Baker (2024) seeks to capture an insight of Panther & Köpcke (1993) (also expressed in Jackendoff & Culicover 2003) that there is a kind of thematic role matching in these examples too, but it does not involve the agent/holder thematic role, but rather the benefactive thematic role. As a result of passive, the PRO subject of ModP is not the agent of a promoting event, but rather its theme-benefactive argument. This induces a raising-like interpretation of the covert modal in the embedded clause rather than a control-like one, such that ModP does not express a holder of the modalized eventuality in this case. Therefore, there is no holder associated with ModP to be equated to the holder expressed in ResP in these examples, and thematic uniqueness of the holder role is not invoked. This vacuum allows another kind of thematic uniqueness to manifest itself. Panther & Köpcke observe that normally the one who benefits from a promising event is the goal, the one who receives the promise, whereas the one who benefits from a persuading event is canonically the agent, the one who does the persuading. So somewhere in the formula for the embedded clause of (18a-c) is a term like “ben(e)=PRO”. Somewhere in the formula for the matrix clause of (18a) is “ben(e)=John.” Therefore, John=PRO in (18a), an instance of object control. In contrast, the formula for the matrix clause in (18b) includes “ben(e)=Mary” (the agent), so Mary=PRO in (18b), a noncanonical instance of subject control. The payoff is in (18c): advising events are different from persuading events in that the advised event is assumed to benefit the advisee, not the adviser. In this respect, *advise* is more like *promise* than like *persuade*. As a result, control does not shift with *advise* the way it does with these other matrix predicates. The advisee is both the holder of the resulting state and the

benefactive of the advising event, so it is the controller regardless of whether control is induced by shared holderhood or by shared benefactive. The details get somewhat complicated (since it is not the resulting state itself that is equated but the event that characterizes the content of that resulting state), and I do not reproduce them here. But this sketch is hopefully enough to show the advantages that thematic uniqueness has for understanding the subtleties of controller choice with triadic verbs within Baker's theory.

The upshot, then, is the hypothesis in (19) about the theory of obligatory control:

(19) X controls Y if and only if $X=Y$ as a result of a thematic uniqueness calculation.

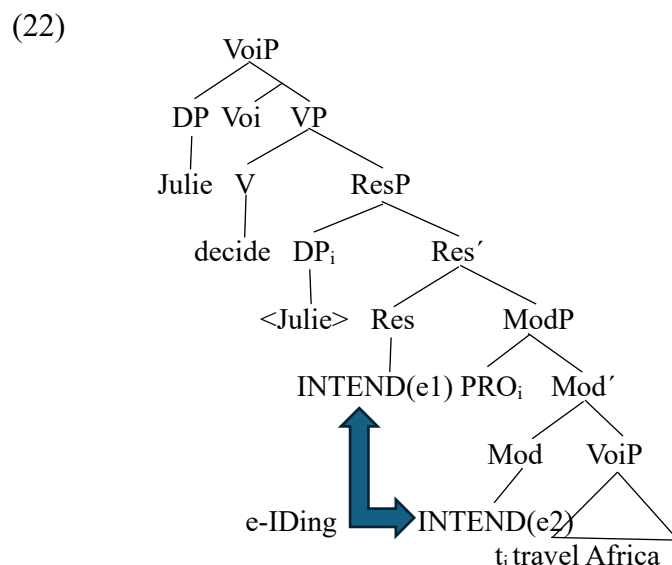
However, Baker (2024) stops short of saying that OC reduces entirely to considerations of thematic uniqueness; rather, he speaks of thematic uniqueness as licensing or inducing OC. This is because there are some reasons to think that OC has a syntactic aspect to it as well. One is that OC seems to be subject to the Phase Impenetrability Condition, which Baker uses to derive the fact that PRO must be in the subject position (at the edge) of the embedded clause (so bad is **Mary promised John_i for the company to promote PRO_i to project manager*; contrast (18a)). There is also some evidence that OC relationships are visible to agreement (Baker 2024) and case transmission (Landau 2008), which hold in the syntax or PF, not just in semantics. This more narrowly syntactic aspect of OC plays a small role in what follows.

3. Why Obligatory Control Fails with Many Propositional Verbs in English

Now let us apply Baker's theory to dyadic verbs rather than triadic verbs, seeing if it can be used to explain additional patterns. Dyadic verbs of the desiderative class allow OC in English, as in (20). In contrast, most dyadic verbs of the propositional class do not allow OC in English, as in (21). One would like to explain why this is the case.

- (20) a. Julie_i decided [PRO_i to travel to Africa in the spring].
 b. Julie_i vowed [PRO_i to drink less alcohol in 2026].
- (21) a. *Julie_i said/declared [PRO_i to have traveled to Africa last fall].
 b. *Julie_i believes [PRO_i to be in excellent health].

For the desiderative verbs, Baker's account of control with triadic verbs like *promise* and *order* carries over easily. There is evidence that there is a modal inside the infinitival complement of these verbs, in that they can have adverbs that refer to future times, as in (20a,b). They are also infelicitous with the kinds of predicates that do not go well with root modal meanings (#*Jennifer decided to be tall tomorrow*, #*Samantha vowed to believe that the sky is green*). And it is plausible to say that these verbs have resulting states, which can be syntactically represented as ResPs. For example, the result of Louise vowing to drink less alcohol is that Louise is obligated to do something, and the result of Julie deciding to go to Africa is that Julie intends to do something. We can then identify the obligation or intention denoted by the ResP in the matrix clause with the obligation or intention denoted by the ModP with a covert Modal head that constitutes the embedded clause. When this is done, the holder of the obligation or intention denoted by the matrix ResP (Julie) is equated with the holder of the obligation or intention denoted by the embedded clause (PRO). Therefore, *Julie* controls PRO, by thematic uniqueness, and OC is licensed. This can be represented as in (22).



In contrast, the crucial ingredients that underlie obligatory subject control with verbs like *decide* and *vow* are notably absent in examples like (21) with *say/declare* and *believe*. On the one hand, when these verbs take an infinitival clause, there is no evidence that it contains a covert modal. For example, the event denoted by the infinitival clause is not in an unrealized future with respect to the time of the eventuality denoted by the matrix clause:⁷

- (23) a. Louise believes [**(herself) to be in excellent health (*in five years)*]
 b. Wendy declared [**(herself) to love her children (*when they become teenagers)*].

Nor are the complements of these verbs restricted to having the sorts of predicates that can be done voluntarily, hence to eventualities that one can intend or be obligated to do. The examples in (24) are felicitous, in contrast to those in (13).

- (24) a. David believes [**(himself) to be tall*].
 b. Christine declared [**(herself) to believe that the sky is green*].

Events of saying, declaring, and believing also do not have the same kinds of resulting states that events of deciding and vowing have. Louise saying, declaring, or believing something does not have much of a notable effect on Louise (or anyone else), such that she acquires a new property at the end of the event (except for trivial properties like those expressed by the present

⁷ Here I illustrate these properties with an ECM infinitival complement rather than a control infinitival complement, because these verbs cannot have a control infinitive; that is what I am trying to explain. The assumption that justifies this is that there is no fundamental difference between ECM complements and control complements when it comes to the structure of the embedded clause, as argued by Wurmbrand (2014): both can have a covert modal, and both can lack one, depending on the matrix verb. I also shift temporarily from *say* to *declare*, because *declare* allows an ECM complement more readily.

perfect—the property of having said something). Overall, then, there is no evidence of a modal eventuality in the infinitival complement or of a resulting state in the matrix clause. Therefore, there are no eventualities to equate that are parallel to the ones that are equated in desiderative sentences like (20). The only eventualities that we have clear evidence for are the saying and the going to Africa in (21a). But these are clearly not the same eventuality. Therefore, thematic uniqueness does not apply. Given that OC relationships need to be licensed by thematic uniqueness, as in (19), it follows that there cannot be OC with ordinary propositional verbs in English.

In short, Baker’s theory of OC is one in which verbs need to have specific lexical semantic properties in order to participate in OC. The flip side of this is that verbs that do not have those specific semantic properties are not expected to participate in OC. Therefore, this is the kind of theory which can explain why some verbs are not control predicates. Most propositional verbs in English are a notable case in point, I claim.

4. Why Obligatory Control Succeeds with *claim* (and a few others)

This line of reasoning then leads us to the question of why OC is possible with the verb *claim* in English, unlike most of the other verbs in the propositional class. Ideally, this would be the proverbial exception that proves the rule. In this section, I argue there is another kind of thematic link between the matrix clause and the infinitival complement that is particular to *claim*, and OC is licensed by that. I claim that *claim* allows for a covert *res* argument, which expresses the individual which the claim is about. Meanwhile, the (nonmodal) infinitival complement can be construed as having content that is about its subject. Inasmuch as they bear a kind of thematic role, these two *about* arguments are subject to thematic uniqueness. The result is OC that

licensed by shared *about* arguments rather than by shared theme arguments, as in SVCs, or shared holder arguments, as with desiderative verbs.

4.1 Initial motivation from the lexical semantics of *claim*

The first step along this path is to motivate that *claim* sentences can have by default a covert *res* argument which is anaphoric to the agent of the claiming—a tacit version of “about X-self”. We can hope to see this by considering how *claim* has different nuances from other verbs that express similar eventualities. These nuances should ideally be discernable apart from control, in sentences where *claim* selects a finite CP or a DP object rather than a nonfinite clause.

Indeed, the propositions that are used as the complement of *claim* are often about the agent of the claiming event. In a Google search for 100 examples of the form “he/she claimed that” (carried out January 24, 2024), 47 out of 87 distinct examples (54%) had a pronoun that could refer to the subject of *claim* inside the CP complement. In comparison, a mere 4 of 87 (5%) had no coreferential pronoun but a pronoun that referred to someone else, suggesting that there is a topic of the embedded clause that is different from the matrix subject. Finally, 27 of the examples (31%) had a CP complement with no human-referring pronoun at all.⁸ An analogous search for 100 examples of the form “he/she thinks that” yielded quite a different pattern: only 19 of 68 distinct examples (28%) had CPs with a potentially coreferential pronoun, whereas 18 of 68 (26%) had a pronoun referring to someone else and 23 (34%) did not have any human-referring pronoun at all. The pattern for “he/she said that” was intermediate between the other two: 34 of 84 (40%) had a coreferential pronoun in the complement clause (more than with *think*

⁸ I tried not to count multiple hits for the same sentence more than once, which is why there are fewer than 100 sentences of each type. This is especially an issue when there is a relevant pop song that the internet refers to often, like “He thinks that I’m an angel” or “She thinks that she’ll marry.” Fortunately, pop songs were not much of an issue for *claim*, as one would expect. There were also examples in which *that* was a demonstrative DP rather than a C head (especially with matrix verb *say*), and others which I counted as not applicable because they were part of metalinguistic discussion like “Is ‘she claimed that’ grammatical in English?”

but less than with *claim*) and 12 of 84 (14%) had a pronoun referring to someone else (more than with *claim*, but less than with *think*). (Only 7 of 84 (8%) had no pronoun, and a whopping 31 were “not applicable”; see note 7.) This suggests that events of thinking are at least as likely to be about someone else as they are to be about the thinker, whereas events of claiming are quite likely to be about the claimer. This is an initial hint that there is a default expectation that claims are about the claimer in a way that is different from other, semantically more neutral verbs.

Of course, claims are not always about the claimer in any direct sense. Some 31% of the googled examples were not, after all. For example, as researchers we are very familiar with making claims about scientific hypotheses, as in (25a).

- (25) a. Martha claimed that protons are made up of gluons and neutrinos.
 b. Martha said that protons are made up of gluons and neutrinos.
 c. Martha admitted that protons are made up of gluons and neutrinos.

We can still ask how a sentence like (25a) with *claim* is a bit different from versions that use other matrix verbs like *admit* ((25c)) or simply *say* ((25b)). (25a) has the sense that protons being made up of gluons and neutrinos (if true) reflects positively on Martha. She has planted her flag on this hypothesis in some way, staking some of her scientific reputation on it. In that sense, even a general law of physics can have to do with the claimer. (25a) with *claim* contrasts with (25c) with *admit* in this respect. (25c) clearly has the sense that Martha was initially invested in the idea that protons are not made up of gluons and neutrinos. This was someone else’s hypothesis, which was borne out by data from a new particle accelerator, and it suggests that Martha now professes the proposition with a shade of reluctance. What then is the opposite of (25c), where Martha has a twinge of happiness and vindication? I claim it is (25a) (although the positive sense of (25a) is perhaps less strong than the negative sense of (25c)). In contrast to both (25a) and

(25c), (25b) with *say* is truly neutral. This does not express Martha having any particular stake in the truth of the scientific hypothesis one way or the other; she may just be repeating something that she heard others say, with little understanding or interest. So *claim* sentences come with some degree of expectation that the claimer has a positive interest in what is claimed. This is related to there being an expectation that the claim is about the claimer in some sense.

This hypothesis that a claimer has a positive interest in what is claimed needs to be understood in relative terms. At first glance, (26a,b) are not about the claimer, nor are they positive things for the claimer.

- (26) a. Mary claimed that John cheated her out of her life savings.
b. Mary claimed that John is an unscrupulous jerk.

Although the complement clause in (26a) reports something negative that happened to Mary, it invites the idea that she now has some right of compensation or retribution with respect to John, which could have advantages for her. In contrast, *?Mary claimed that she cheated John out of his life savings*, is a bit off; although the CP complement is about Mary, she does not have a positive stake in it, and a matrix verb like *admit* or *confess* would be more comfortable. Similarly, (26b) can be understood as denigrating John in a fashion that builds up Mary's standing relative to John in some way.

These nuances of the verb *claim* can also be seen when it takes a DP object. Thus, (27a) has a sense that Mary is entitled to the particular right or the winnings, and that taking hold of them is likely to benefit her. In contrast, *Mary claimed her gambling debts* is a bit off; one prefers a verb like *acknowledged* or *admitted*. (27b) is instructive in that responsibility for something can be either a positive thing or a negative thing, depending on whether the event is

praiseworthy and subject to reward, or blameworthy and subject to punishment. Using *claim* suggests that (Sarah thinks) it is praiseworthy in this case.⁹

- (27) a. Carl claimed his right to a new trial/the grand prize in the lottery.
b. Sarah claimed responsibility for what happened.

This reinforces what we discerned using finite CP complements in (25) and (26).

As a further linguistic property relevant to this topic, I believe that *claim* in English differs from other propositional verbs like *say* and *think/believe* in that it resists having non-*de-se* readings of pronouns in its finite CP complement. Consider a classic case of unrecognized identity like the following. Frank is the judge of a singing contest similar to American Idol, but in which the contestants are anonymous, known only to the judge(s) by recordings of their songs, each of which is assigned a number. Unbeknownst to Frank, his friend Susan submits a recording of Frank singing, labeled as Singer #8. Frank does not recognize his own voice, but he falls in love with recording #8. He thinks “#8 is the best” and he records his decision that “#8 is the winner.” In this case, verbs like *think* and *say* famously allow a non-*de-se* reading of the pronoun ‘he’ referring to Frank, as in (28a,b) (Chierchia 1989 and many others). This also carries over to less common verbs of communication like *declare* and *announce*, which also make sense in this situation as constructed.

- (28) a. Frank thinks that he is the best singer in the contest this year.
b. Frank said/told everyone that he was the best singer in the contest this year.
c. Frank announced/declared that he was the winner of the contest this year.

⁹ Of course, someone can claim responsibility for something that most people consider a negative thing, as in *The terrorists claimed responsibility for the explosion that killed 20 children*. But in the warped worldview of the terrorists, the explosion was a positive thing, in that it adds to their reputation as not to be trifled with and advances their political objectives.

However, my consultants and I find that a similar sentence with *claim* is notably degraded with this kind of non-*de-se* interpretation for a pronoun in its CP complement. (29) is less felicitous in this context than (28a-c).¹⁰

(29) ?#Frank claimed that he was the winner of this year's contest.

Example (29) is possible, of course, but only if Frank knows that it is he himself who he is asserting to be the lucky winner. Although this point of variation among the CP-selecting verbs of English has not been noticed before as far as I know, it is subtle but consistent for the speakers I have consulted. And I believe that it is an important hint as to why *claim* allows OC but *think*, *say*, *announce*, and *declare* do not. It is plausible that there is a connection, since OC constructions do not allow non-*de-se* readings either (Chierchia 1989). *Claim*'s affinity for *de se* readings, observable in (29), might then have something to do with why it allows OC. (There are other verbs in English that I find to be like *claim* in this respect, including *admit*, *confess*, *affirm*, *deny*, *boast*, and *brag*; see below for discussion.)

4.2 OC induced by aboutness

How then can we capitalize on this sense that OC with *claim* in English depends on the *res* thematic role in a way that is somewhat analogous to how OC with *promise* depends on the holder role or the benefactee role?

Consider first the relationship of an explicit *about* phrase to the predicate *claim*. These can cooccur, both with *claim* in its verbal guise ((30a)) and with the related noun ((30b,c)).

(30) a. Mary claimed about John that he is an unscrupulous jerk.

¹⁰ This is the judgment of myself and four other native speakers of English (two linguists, and two not). For me the judgment is clearer when I imagine that the choice to describe the event as a claiming rather than as a saying or thinking or declaring or announcing resides with the speaker, rather than with Frank. In other words, (29) feels wrong when what Frank actually said is something like “#8 is the winner of this year's contest” but may be somewhat better if Frank himself says “I claim that #8 is the winner of this year's contest.” (I have not needed to explain this to my consultants for them to judge that (29) is worse than (28a-c), however.)

- b. Mary's claim about John was that he is an unscrupulous jerk.
- c. Mary's claim was about John.

Moulton (2009) and Rawlins (2013) discuss some of the issues surrounding *about* phrases, fitting them into the framework for analyzing propositional attitude verbs developed by Kratzer (2006), Moulton (2009, 2015), Elliott (2017), and (somewhat independently) Moltmann (2024). Moulton takes the *about* phrase to be the expression of a *res* argument of a propositional attitude verb—a kind of argument that such verbs have but most other verbs do not. Rawlins claims that the *about* phrase is a predicate that can be predicated of the content-bearing entity that is associated with an event of (say) claiming (cf. (30c)). I take this content-bearing entity to be the claim that is produced by the claiming event, following Kratzer, Moulton, and Moltmann's version (in contrast to Elliott (2017), who assumes that the event itself has content). Rawlins contrasts his view to Moulton's, but I assume that they bring out two aspects of what is essentially the same thing. I take it that "X is an argument of Y" implies that "X bears a thematic role with respect to the event denoted by Y", thematic roles being predicates of events in neo-Davidsonian fashion. In that sense, being an argument of a verb and being a predicate of an event (or something closely related to it) are systematically related. (30a) then corresponds to the semantic formula in (31).

- (31) $\lambda e \exists z \text{ claim}(e) \ \& \ \text{agent}(e)=\text{Mary} \ \& \ \text{theme}(e)=z \ \& \ \text{about}(z)=\text{John} \ \& \ \text{content}(z)=\text{he is an unscrupulous jerk.}$

Crucially, *about* phrases are like canonical arguments in that they are unique. A clause headed by a verb like *claim* can have only one *about* phrase, so (32a) is bad (cf. Paillé 2024: 16). (Contrast this with a conjoined DP inside a single *about* phrase, which is fine, as in (32b)).

- (32) a. *Tom claimed about Sue about Bill that she had a crush on him.

- b. Tom claimed about Sue and Bill that they are in love.

About arguments behave in this respect like instrumental phrases and benefactive phrases, which are also optional, are implied by certain verbs but not others, and must be unique in a clause (**Mary fixed the sink with a screwdriver with a hammer*; **John baked a wedding cake for Mary for Sam*; see Paille 2024: 5). Therefore, it makes sense to treat the *res* argument as being subject to the thematic uniqueness constraint, parallel to other more familiar arguments.¹¹

Now my hypothesis about what makes *claim* different from most other verbs of cognition or communication in English is that it can have a null anaphoric expression of an *about* phrase in its semantic representation. So (33a) is naturally understood similarly to (33b).¹²

- (33) a. Frank claimed that he was the best singer in this year's singing contest.

- b. Frank claimed about himself that he was the best signer in this year's singing contest.

The denotation for the matrix Voice' of (33a) would then be (34). And a formula for a version with an infinitival complement would be similar, with perhaps some adjustments internal to the "content(z)=P" term.¹³

¹¹ This would be formally more similar if it is the claiming event itself that has content, as in Elliott (2017), rather than its (often covert) theme argument. Then *about* would be a function from an event to an entity, just as the familiar thematic roles of agent, theme, and goal are in the neo-Davidsonian approach. However, I do not think this is quite accurate; see Moltmann (2024) for relevant discussion. Perhaps then *about* phrases are thematic roles in a somewhat extended sense. Compare Moulton's claim that the content of a content-bearing entity is unique. This accounts for why a verb or noun like *claim* can only cooccur with a single finite CP, as in classical generative syntax, even though the CP is technically a predicate rather than an argument in the Kratzer-Moulton analysis.

¹² One can compare *claim* with verbs of bodily care like *wash* and *dress* in this respect. It is well-known that these can have a theme argument distinct from the agent (*Sam washed/dressed Fred*), but they can also not have an overt theme argument. When they do not, the theme argument is interpreted like a reflexive bound by the agent (*Sam washed/dressed*, similar in meaning to *Sam washed/dressed himself*). Other optionally transitive verbs like *eat* are not "naturally reflexive" in this way. My hypothesis is that *claim* is "naturally reflexive", rather like *wash* is, but with regard to its *res* argument rather than its theme argument.

I leave it open whether there is a covert representation of an anaphoric *res* argument in the syntax of a sentence like (33a), or whether the term [*& about(z)=x*] in (34) is simply added in the semantics. This does not matter for the calculation based on (19) that leads to OC. Different cases probably work differently. I tend to think that the null theme argument in an SVC is syntactically represented at least sometimes (Collins 1997), as is the PRO in an infinitival complement. On the other hand, the holder of the modal eventuality denoted by a ResP seems not to be syntactically represented, as discussed by Baker (2024). The *res* argument of *claim* could go either way.

¹³ The idea that *claim* by default has a *res* argument that is bound by the agent of *claim* should also provide a basis for accounting for the observation that, in examples like (29), a coreferential pronoun in the CP complement of

(34) $\lambda x \lambda e \exists z \text{ claim}(e) \ \& \ \text{agent}(e)=x \ \& \ \text{theme}(e)=z \ [\& \ \text{about}(z)=x] \ \& \ \text{content}(z)=\text{he is the best singer this year.}$

We also need to consider aboutness relations that are expressed inside the infinitival complement. Does the complement have a thematic role that can link up with one in the matrix clause in the right way to invoke thematic uniqueness, thereby inducing OC? If what is special about claims is that they tend to be about the claimer, then we should consider what the infinitival complement clause is naturally about. I assume the simple generalization in (35).

(35) A clause can, by default, be interpreted as being about its subject.

In many cases, (35) is obvious and presumably uncontroversial; for example, a sentence like *Steve is tall* is naturally construed as being about Steve. So is a sentence like *Steve is proud of Karen*, even though in this case another individual is named inside the predicate. So is a sentence like *Steve likes Karen*. Such a sentence might be able to have other aboutness construals, depending on its context and its intonation, but that it is a statement about Steve is surely one of them. Indeed, many functionalist approaches take it as a truism that there is a close, nonaccidental relationship between subjecthood and topicness. Rizzi (2005, 2006) is one who has developed a version of this within Chomskyan syntax, arguing that an aboutness relationship holds between a DP in the preverbal subject position and the rest of the sentence, such that that DP “is selected as being the point of departure in the description of the event, which is described as being about that [DP].”¹⁴ He also shows that this has consequences for discourse, affecting the

claim strongly favors a *de se* reading. Compare Moulton (2009: 110), who shows how the matrix subject binding a covert *res* argument of *believe* leads naturally to a *de se* interpretation of a covalued pronoun in the complement clause given his semantics. I’m sure that there are niceties about this to work out, but I do not attempt that here.

¹⁴ The notion of aboutness that Rizzi has in mind may be somewhat different from the one that is involved in explicating the semantics of the *res* argument of an attitude verb, which can show up as a PP headed by *about*. In particular, it may not be true that there is only a single DP/entity that a given clause/proposition is about for Rizzi. He assumes that both preverbal subjects and clitic-left-dislocated topics bear a similar aboutness relationship to the clause they have scope over (whereas they differ in their relationship to the prior discourse: topics must be “D-linked” whereas subjects need not be). However, an Italian sentence can have both a preverbal subject and a topic, or

interpretation of a null pro in the following clause. (See Cardinaletti (2004: 151) for a related idea.)

This gives us the ingredients that we need to analyze OC in an example like (36) as being induced by thematic uniqueness. Its core semantic value is (37a). (37) extends this by adding that the claim is by default about the claimer. (37c) further extends it by adding the consequence that clauses are by default about their subject.

(36) Mary claimed [PRO to have paid the fine].

(37) a. $\lambda x \lambda e \exists z \text{ claim}(e) \ \& \ \text{agent}(e)=\text{Mary} \ \& \ \text{theme}(e)=z \ \& \ \text{content}(z)=[\text{PRO paid the fine}]$

b. $\lambda x \lambda e \exists z \text{ claim}(e) \ \& \ \text{agent}(e)=\text{Mary} \ \& \ \text{theme}(e)=z \ \& \ \text{about}(z)=\text{Mary} \ \& \ \text{content}(z)=[\text{PRO has paid the fine}]$

c. $\lambda x \lambda e \exists z \text{ claim}(e) \ \& \ \text{agent}(e)=\text{Mary} \ \& \ \text{theme}(e)=z \ \& \ \text{about}(z)=\text{Mary} \ \& \ [\text{content}(z)=[\text{PRO has paid the fine}] \ \& \ \text{about}(z)=\text{PRO}]$

Therefore Mary=PRO

The result is that there are two statements that express what the claim (produced by Mary's action) is about: one derived from the particular lexical semantics of *claim*, the matrix verb, and the other derived from the embedded clause based on how it is structured as subject and predicate. One of these says that the claim is about Mary; the other says that the claim is about PRO. But it is the same claim. This follows because the CP is a predicate of the theme of the claiming in the Krazter-Moulton theory, as is the covert *about* phrase. Finally, *about* is a thematic role predicate, subject to thematic uniqueness. Therefore, PRO must be equated with Mary. This licenses the OC of PRO by *Mary* with the verb *claim*.

two or more CLLD topics. But even if the two senses of aboutness are not identical, it is very plausible that they are related. (35) can be read as saying that something that the clause is about in Rizzi's sense can also have the *about* thematic role—aboutness in a stronger sense that licenses uniqueness conclusions.

This then is the core of my account. I claim that it is adequately similar to Baker's account of OC with desiderative verbs like *promise*, as long as one accepts the idea that aboutness is a thematic role, on a par (more or less) with agent and benefactee. Like the other thematic roles, this one can be overlaid on other thematic roles but it is nevertheless subject to thematic uniqueness.

I acknowledged above that claims are not always about the claimer. This is a defeasible expectation, which can be overridden rather easily—for example, by including an overt *about* phrase with an object not bound by the subject of the clause (see (30a)). However, a prediction that emerges is that including an overt *about* phrase will interfere with the possibility of OC, since it takes away the possibility of aboutness relationships connecting the matrix subject with the embedded subject. This is true, as seen in (38). (38a,b) show that the pronoun subject of a finite CP complement of *claim* can refer to the matrix subject. This is not disrupted by the possibility of including an *about* phrase whose object refers to someone else mentioned in the embedded clause (here Mary). Thus, ordinary pronominal coreference is not very sensitive to aboutness relations—as we expect, given that that is a relationship that is subject to few if any grammatical conditions. (38c) is parallel to (38a), except that the complement clause is an infinitival and its subject is PRO rather than an overt pronoun. It is fine. However, (38d) shows that including an *about* phrase referring to Mary does disrupt *John* controlling PRO in this case. As a result, (38d) is significantly degraded compared to both (38b) and (38c).

- (38)
- a. John_i claimed that he_i was the one who discovered Mary and made her a star.
 - b. John_i claimed about Mary_k that he_i was the one who discovered her_k and made her_k a star.
 - c. John_i claimed [PRO_i to have discovered Mary and made her a star].
 - d. ??John_i claimed about Mary_k [PRO_i to have discovered her_k and made her_k a star].

This confirms that OC with the propositional verb *claim* crucially depends on aboutness relationships, supporting my proposal.¹⁵

There is a significant fact that is not covered by this theory. My analysis as it stands predicts that an *about* phrase in the matrix clause with a verb like *claim* could itself control PRO in the embedded clause. It finds nothing wrong with (39a,b), although I find them unacceptable.

(39) a. ?*Lucy claimed about David_i [PRO_i to have embarrassed himself_i at the party].

b. ?*Jerry claimed about Joy_i [PRO_i to be the most talented singer that he has heard].

I tentatively assume that (39a,b) violate a more strictly syntactic condition on control, such that the OC controller must c-command PRO. It was mentioned above that Baker (2024) argues that OC involves establishing a syntactic relationship, which is subject to the PIC and visible to morphosyntactic processes like agreement and case concord. This is over and above the fact that OC relationships need to be interpreted by way of thematic uniqueness. Given this, it would not be much of a surprise if the syntactic relationship of OC requires that the controller c-command the controllee as well. The examples in (39) violate this further condition, on the assumption that *about* is a true P, heading a PP in the syntax. This requires distinguishing *about* from “light” functional Ps like *to* and even *on*, which can be analyzed as being case-markers (dative, locative case) rather than true Ps in the syntactic representation, given that the objects of *to* and *on* can be OC controllers, as in (40) (cf. Landau 2015: 15).

(40) a. Catherine signaled to Nick_i [PRO_i to turn right].

b. Julia depended on Lynn_i [PRO_i to bring a salad to the dinner party].

¹⁵ This example was also checked with four consultants other than the author, two linguists and two nonlinguists. My account predicts that ?*John claimed about himself to have discovered Mary* should be possible, with the overt *about* phrase not disrupting subject control if it is coreferential with the subject. I find the judgment to be somewhat intermediate: it is more awkward than (38c) but not as bad as (38d). I assume that it is basically grammatical, but dispreferred because the overt *about* phrase adds to the length and prosodic complexity of the sentence without contributing anything much to its meaning.

This is not an obvious distinction in English, by any means, but it is a familiar kind of distinction to make, at least for *to*. How precisely to implement it, I leave to future research.¹⁶

4.3 Further consequences

This analysis has some additional positive consequences. At the top of the list is that it explains why there is no control shift with the propositional verb *claim*. I observed in section 1 that *claim* can have an indirect object, but even when it does, it remains strictly a subject control verb. The indirect object cannot be the controller, as shown again in (41).

- (41) a. Ben_i claimed to the judge [PRO_i to have paid the fine].
b. *Ben claimed to the judge_i [PRO_i to have treated him unfairly].

This is a bit surprising for theories that emphasize the rarity or markedness of subject control, such as those that include some version of the Minimal Distance Principle or invoke a thematic hierarchy. *Claim* seems to be different in this respect from a verb like *ask*, which has subject control when it has no object, but often shifts to object control when it has an object (although *ask* also allows subject control in the presence of an object, when circumstances are favorable).

- (42) a. Diane_i asked [PRO_i to leave work early].
b. Diane asked Richard_i [PRO_i to leave work early].

My theory easily accounts for this observation. It asserts that OC with *claim* hinges on the *about* role, rather than on agent or benefactive-goal roles. Including a DP with a goal role is independent of this and therefore has no effect on it. The semantic representation for (41a) is

¹⁶ This proposal also rules out control by an *about* phrase with matrix verbs other than *claim*, which have fewer lexical semantic conditions, and do not otherwise allow subject control, such as **Jeff said/believes about Ann_i [PRO_i to be a beautiful singer]*—a fact that my theory otherwise would not account for.

simply (43), minimally different from the one in (37c). The term “goal(e)=the judge” has been added, but it is inert with respect the thematic uniqueness relationships that induce OC.¹⁷

(43) $\lambda e \exists z \text{ claim}(e) \ \& \ \text{agent}(e)=\text{Ben} \ \& \ \text{theme}(e)=z \ \& \ \text{goal}(e)=\text{the judge} \ \& \ \text{about}(z)=\text{Ben} \ \& \ [\text{content}(z)=[\text{PRO has paid the fine}] \ \& \ \text{about}(z)=\text{PRO}].$ Therefore Ben=PRO

This was the third major puzzle about OC with propositional verbs, and now we have an account of it as well, at least in the special case of *claim* in English.

An important design feature of Baker’s theory of controller choice for desiderative verbs is that it also accounts for the control shift which can happen when the embedded clause is nonactive. Simplifying some, when the subject of the embedded clause is a goal-benefactive rather an agent, it can be controlled by the goal-benefactive object of ‘promise’ rather than by its agent-subject argument, as seen in (44b). This is because thematic role matching is at work.

(44) a. John_i promised Mary [PRO_i to finally give her her due at the shareholders’ meeting].
b. John promised Mary_i [PRO_i to finally be given her due at the shareholders’ meeting].

In contrast, having a passive (or other non-agentive) verb rather than an active verb in the infinitival clause of *claim* has no effect on controller choice. (45) can only be understood as having subject/agent control, no different from (41b).

(45) Hannah_k claimed to the judge_i [PRO_{k,*i} to have been given t_{k,*i} an exemption/a bribe].

This too is a straightforward consequence of the fact that OC with *claim* rides on the *res* thematic role rather than on the agent or goal/benefactive role. Hannah is understood as being the *about* argument of *claim*, as before. PRO can be understood as the *about* argument associated with the infinitival clause, given (35). Therefore, there is subject control and not object control in (45) as

¹⁷ See below for some discussion of whether the content of a communication can ever be assumed to be about the goal of the communication rather than the agent of the communication. (There are a few possible cases in French, but it seems to be quite rare/marked; see fn.25 for some discussion.)

well. The fact that *Hannah* is an agent as well as the *res* in the matrix clause does not come into it, nor does the fact that PRO is a goal as well as a *res* argument related to the embedded clause.

- (46) $\lambda e \exists z \text{ claim}(e) \ \& \ \text{agent}(e)=\text{Hannah} \ \& \ \text{theme}(e)=z \ \& \ \text{goal}(e)=\text{the.judge} \ \& \ \text{about}(z)=\text{Hannah} \ \& \ [\text{content}(z)=[\text{PRO was given an exemption}] \ \& \ \text{about}(z)=\text{PRO}].$

Therefore Hannah=PRO

Within this framework, a nominal expression can bear more than one thematic role. Thematic uniqueness does not require that every thematic role associated with one argument be identified with every thematic role associated with another argument, only that matching roles must match. We thus derive the right result here too, explaining another way in which OC with the propositional verb *claim* behaves differently from OC with desiderative verbs like *promise*.

Here is another way that *claim* differs from control with desiderative verbs—one not noticed in previous literature, as far as I know. Dyadic verbs like *decide* in English can have impersonal passives. When they do, the implicit agent is still understood as controlling PRO in the infinitival complement. As a result, (47b) is quite similar in meaning to (47a).

- (47) a. They_i decided [PRO_i to close the store after the holiday season].
b. It was decided (by the management_i) [PRO_i to close the store after the holidays].

The possibility of control by an implicit agent in sentences like (47b) has sometimes been thought to be a rather sporadic phenomenon, but van Urk (2013) shows that it is quite systematic across Germanic languages, with the qualification that a passive agent cannot control PRO when the matrix verb selects an internal argument that T agrees with in the passive version. This agreement is what prevents agent control with the passive of a verb like *promise*, a well-known observation sometimes called Visser's Generalization. Thus, examples like *It is widely hoped to have enough money to retire* are also reasonably acceptable, all things being equal. Now, *claim*

also allows an impersonal passive when it takes a finite CP complement as in (48a). However, an impersonal passive with *claim* selecting a control infinitive is strikingly bad, in sharp contrast to (47b). (48c) is not possible as a near-equivalent of (48b), for example.

- (48) a. It is often claimed [that Patrick/one is innocent].
 b. (Lawyers know that) people_i accused of a crime often claim [PRO_i to be innocent].
 c. (Lawyers know that) *it is often claimed [PRO to be innocent (when accused of a crime)].

This can be explained by saying that there is a clash between suppressing/backgrounding the agent of the clause by passivization while at the same time foregrounding it by identifying it as the topic of the content associated with the event. This can be seen apart from OC by the weirdness or artificiality of (49a), where the implicit agent of the passive binds an anaphor inside the *about* phrase. ((49a) contrasts with (49b), which shows that implicit agents can bind indefinite anaphors in other kinds of PPs.)

- (49) a. ??It is often claimed about oneself that one is innocent.
 b. Such privileges should be kept to oneself. (Baker, Johnson et al. 1989: 228)

On my theory, the badness of (48c) reduces to the badness of (49a), attributed to the pragmatic clash of a null indefinite agent binding an anaphoric *about* argument. (48c) has OC of PRO in the infinitival complement. This OC depends on there being the equivalent of a null anaphoric *about* phrase in the matrix clause, licensed by the lexical semantics of *claim*. But this anaphoric element cannot be bound by a backgrounded, indefinite implicit agent argument, as in (49a). Therefore, (48c) is unacceptable. *Claim* is different from *decide* in this respect, because it depends on shared aboutness relationships between the matrix clause and the embedded clause.

Another positive consequence of this theory is that it can give us ideas about where to look for other propositional verbs in English which might also allow OC. The received wisdom is that there may not be any others besides *claim* (and perhaps *pretend*). However, my diagnosis

of what it is about *claim* that makes it different from verbs like *say* and *believe* is that it comes with a default expectation that the content of its CP complement is about the agent of the event.

There is no reason to expect that *claim* is the only verb in English that comes with such an expectation. On the contrary, I realized that claims generally reflect well on the claimer by contrasting sentences with *claim* to sentences with verbs like *admit* and *confess*, for which the content of the complement clause typically reflects badly on the agent of the event, as in (50).

- (50) a. George claimed/?admitted/?confessed that he saved the day for his team.
b. George admitted/confessed/?claimed that he ruined the game for his team.

Perhaps even more clearly, boasts and brags need to be about the boaster/bragger.¹⁸

- (51) Rebecca boasted/bragged that she/her daughter/?Mary saved the day for the team.

A way of sharpening this intuition is to see if these verbs allow a pronoun in CP their complement to have a non *de se* reading. My judgment is that they are like *claim* and different from *say* and *think* in resisting such a reading. Imagine again the scenario in which Frank is the judge of an anonymous singing contest, and Toni submitted a recording of him singing which Frank does not recognize but thinks is great/terrible. Not good in this context are (52a,b).¹⁹

- (52) a. #Frank admitted/confessed that he was the worst singer in the contest this year.
b. #Frank boasted/bragged that he was the best singer in the contest this year.

Also resistant to the non *de se* interpretation for what feel like similar reasons are *affirm*, *acknowledge*, and their antonym *deny*.

¹⁸ I thank Sreekar Roghatham Matam for pointing out these verbs to me. There is no necessary syntactic link between a DP in the embedded clause and the matrix subject: (51) is fine if we know that Mary is Rebecca's daughter, for example. It is not the case, then, that the CP complement must contain a pronoun bound by *Rebecca*. However, it is odd if Rebecca has no close relationship with Mary, such that she identifies with her in some way.

¹⁹ For *admit*, I tested this judgment with four consultants, in comparison with *think*, *declare* and *claim*. For the others, it is my judgment alone. See fn.10 for a remark on how (not) to get this judgment; it is less clear of what Frank said was "I acknowledge/deny that Singer #8 won the contest this year."

- (53) a. #Frank affirmed/acknowledged that he won the singing contest this year. (What he actually said is “Singer #8 is this year’s winner”, not knowing that he is singer #8)
- b. #Frank denied that he won the singing contest this year. (He said: “Contrary to the rumors going around, singer #8 did not win the contest this year.)

Therefore, there is reason to say that these verbs also come with the default assumption that the content of the event is about the agent of the event. If that is so, then these verbs should have the link that they need to license OC in an embedded clause. And in fact they do. The reason why this has not been recognized by those who say that *claim* is unique among the propositional verbs in English is that the nonfinite complements of these verbs usually have gerund forms with *-ing* rather than *to* infinitives. In some cases, a preposition like *to* or *about* is needed as well.

However, all the examples in (54) have a null subject in the nonfinite clause that is covalued with the matrix subject in what looks like an instance of OC.²⁰

- (54) a. Mary admitted having (*to have) stolen the money.
- b. John confessed to having (?to have) stolen the money.
- c. Sue bragged/boasted about having (*to have) scored the winning goal.
- d. Karen affirmed/acknowledged having (??to have) gotten lost going to the meeting.
- e. Nancy denied having (?*to have) used a racial slur in her class on Thursday.

So this wider range of propositional verbs can undergo OC, as predicted; the only difference is that their nonfinite clauses are headed by *-ing* rather than by *to*. This is still some kind of

²⁰ This claim is hedged because, unlike *to*-infinitives, gerundive clauses in complement position sometimes allow a sort of non-obligatory control. An example is (i) (cf. Sag & Pollard 1991).

(i) John_i and Mary_k discussed [PRO_{i,k,i+k,arb} running for president of the HOA].

Jackendoff & Culicover (2003) propose that this is a different category of “nearly free control” with some special properties. Landau (2013: 44-45) conjectures that it could be nonobligatory control, made possible by the complement of *discuss* being a nominal gerund, with an extra DP projection. I believe that the examples in (54) do not have the special properties of (i) (allowing a PRO_{arb}), but I do not go through the steps to prove this.

difference with *claim*, to be sure, and one would like to explain it, if possible. Presumably it has something to do with subtle aspectual differences in the complement clause and how it relates to the matrix verb (cf. the subtle difference between *Mary continued to read the book* and *Mary continued reading the book*). At the same time, it is not the case that all propositional verbs in English allow control as long as the complement is a gerund; many do not, as shown in (55).

- (55) a. *John believes having saved the day for the team.
b. *Mary thinks being the best candidate for the job.
c. *Helen said having scored well on the entrance exams.
d. *Sally declared being willing to serve on the admissions committee.
e. *Claire announced having resigned from her job as chair.

So there is still a distinction between propositional verbs that license OC and ones that do not in English. It is just that the former class is not as small as has been thought, once one includes verbs that take *-ing* complements as well as ones that take *to* infinitives. There are several difficult and probably intertwined questions about *-ing* complementation, including what is the semantic difference between them and infinitival clauses, whether they are of category CP or DP, and whether they always involve true OC or something a little bit different (see fn. 20). I do not take up these questions here, keeping the focus on ordinary infinitival clauses. However, it is to the credit of my proposal about what licenses OC with *claim* that it extends in this way. More generally, it supports the hypothesis that there needs to be something about the thematic analysis of the matrix clause that overlaps with that of the embedded clause in order to license OC via thematic uniqueness.

5. Generalized OC with propositional verbs in French

This brings us to the third puzzle about control with propositional verbs that was outlined in section 1: what is the crosslinguistic difference between (say) French and English, such that many more propositional verbs allow OC in French? This difference is not as black-and-white as one might think, since it is due in part to the fact that French does not have English's distinction between infinitival complement clauses and gerundival complement clauses. As a result, analogs of the examples in (54) using the verbs *admettre* 'admit', *avouer* 'confess', *se vanter* 'brag', *affirmer* 'affirm', and *nier* 'deny' are robustly attested with infinitival complements and OC meanings.²¹ Part of the difference between the languages, then, is simply that English draws a morphological distinction between two kinds of nonfinite complements which French does not.²² However, this is not the only difference between French and English in this domain. The French analogs of (55) with infinitival complements are also robustly attested; see also (3b) for *croire* 'believe' and (3c) and (60a) for *dire* 'say'.

- (56) a. Je crois avoir vu Jesus dans mon rêve. (All from the internet)
 'I believe to.have (that I have) seen Jesus in my dream.'
- b. Avec plus d'expérience, je pense avoir un avantage sur le marché du travail.
 'With more experience, I think to.have (that I have) an advantage on the job market.'
- c. J'ai déclaré avoir perdu mon telephone, mais je l'ai retrouvé peu de temps apres.

²¹ By "robustly attested", I mean that either a google search yielded dozens of hits, many of which have propositional OC meanings, or that all four of my consultants accepted them (or both). By "marginally attested", I mean that a google search yielded a handful of hits, some of which have propositional meanings, or that two of my consultants accepted them. Google searches were done by typing in "*j'ai VERB avoir*" and "*j'ai VERB être*" and sorting through the hits with some help from Google translate. (The tense of the main verb could vary between present and passé composé depending on which seemed more natural—e.g., on whether the verb is stative or not.) Of these five verbs, *se vanter* had the fewest hits, on the order of 100; the others all had more than 200. To recognize OC meanings, the most systematic thing that needs to be filtered out is "exceptional ECM" examples, in which the subject of the infinitival clause has been extracted, like in English *the person who I believe – to be in charge* (see Kayne (1984) and Rizzi (1982) for discussion of this possibility in French and Italian). There are also plenty of other random things on the internet.

²² I thank Viviane Deprez (p.c.) for discussion of this point. Note that French has a gerundival verb form, but it is only used in adjunct-modifier clauses, not in complement clauses.

‘I have reported to.have (that I) lost my phone, but I found it a short time after.’

d. La semaine dernière, j’ai annoncé avoir fait +35 000€ de ventes en 10 jours.

‘The last week, I have announced to.have (that I have) made 35k€ in sales in 10 days.’

Indeed, there might be very few verbs of cognition or communication that strictly forbid OC in French. Some are certainly more common than others. In addition to those above, OC is robustly attested for *savoir* ‘know’, *révéler* ‘reveal’, *apprendre* ‘learn’, *expliquer* ‘explain’, *supposer* ‘suppose’, *réaliser* ‘realize’, *comprendre* ‘understand, figure out’, *répondre* ‘answer’, *soupçonner* ‘suspect’, *présumer* ‘presume’, *douter* ‘doubt’, *se plaindre* ‘complain’, *parier* ‘bet’, *informer* ‘inform’, *écrire* ‘write’, *juger* ‘judge’, *préciser* ‘specify’, *oublier* ‘forget’, *jurer* ‘swear’, *considérer* ‘consider’, *garantir* ‘guarantee’, *faire semblant* ‘pretend’, and *imaginer* ‘imagine’.²³

OC is also marginally attested for *entendre* ‘hear’, *vérifier* ‘verify’, *crier* ‘shout’, *déduire* ‘deduce’, *proclamer* ‘proclaim’, *spécifier* ‘specify’, *répliquer* ‘reply’, *stipuler* ‘stipulate’, *insinuer* ‘insinuate’, and *suggérer* ‘suggest’. Three verbs that I did not get internet hits on were the manner speaking verbs ‘mutter’, ‘grumble’ and ‘whisper’, but still two out of my four consultants accepted them to some degree.

(57) a. J’ai chochoté avoir fait une erreur. (three ?, one *)

‘I-have whispered to.have (that I have) made a mistake.’

b. J’ai grommelé/marmonné avoir un problème. (one ok, one ? two *)

‘I-have grumbled/muttered to.have (that I have) a problem.’

²³ The idea to try some of these verbs came from Reed’s (2014) list of verbs that do not allow OC with a bare eventive verb in French. In fact, almost all of them do allow OC with ‘have’+V or ‘be’+Verb (i.e. a stative verb) in the CP complement. Other verbs came from Landau’s (2015) list of canonical propositional verbs, Godard’s (1988) list of verbs that can appear in the gapless *dont* construction (see below), and my own curiosity.

So there are some gradations and perhaps some restrictions, but the OC construction is clearly very productive with propositional verbs in French, and there is a difference between French and English in this regard.²⁴ Why should this be?

At first glance, the wide range of verbs that allow this kind of OC in French may suggest that the difference between the two languages must reside in the infinitival clause rather than in the semantic associations of the matrix verb. However, there is also a sense in which the wide range of verbs undermines that idea. If French allowed non-English-like OC only with positive epistemic verbs like ‘believe’, ‘think’, and ‘know’, one might try to develop the idea that French allows null epistemic modals in an infinitival clause, whereas English only allows null deontics and other root modals. An epistemic modal might overlap in meaning with an epistemic main verb, with the result that OC can happen with epistemic verbs in French in a way that they cannot in English. But these canonical cases of propositional OC in French are far from the only ones. This sort of OC is also possible with main verbs like *douter* ‘doubt’, *deviner* ‘guess’ and *mentir* ‘lie’, in which the subject of the main verb explicitly does not know or does not believe the content of the infinitival complement, as shown in (58).

- (58) a. Je doute être la seule person à faire ça. (internet, robustly attested)
 ‘I doubt to.be (that I am) the only person to do that.’
 b. J’ai deviné avoir un bébé garçon ... Alors, qu’en pensez-vous? (marginally attested)
 ‘I have guessed to.have (that I am having) a baby boy. ... What do you think?’
 c. Il a menti avoir gagné à la loterie, parce qu’il voulait apparaître à la télévision.
 ‘He has lied to.have (that he has) won the lottery because he wanted to.appear on television.’

²⁴Two other verbs that gave no hits were *conjecturer* ‘guess’ and *alleguer* ‘allege’. See below for a comment.

In addition, many verbs of communication, such as *dire* ‘say’, are neutral as to whether the subject of the matrix verb knows or believes the content of its infinitival complement. Therefore, I do not see how a substantive epistemic modal meaning could be given to infinitival clauses in French that would be compatible with the very wide range of matrix verbs they can be used with.

Instead, I make the initially-peculiar-sounding claim that many French verbs can have a slightly different lexical meaning than their English analogs have. Most cognition and communication verbs in French come with the option of grammatically representing that the content associated with the event named by the verb is about the agent or holder of the event—as only *claim* and a few others (*boast, deny*) do in English. In other words, the French verbs all allow an optional “about(*z*)=*x*” term in their semantic denotation, where the variable *x* is also associated with the subject argument of the clause (see (37)). It is not a very familiar idea that languages can differ systematically in the lexical semantics of a large class of verbs in this way. But it is not unheard of either. My proposal is somewhat analogous to Talmy’s (1975, 1985) famous observation that manner of motion verbs in English can take a path expression, as in *Mary ran/swam/danced/waltzed/paraded/skated/... out of the room*, whereas Romance languages for the most part do not allow this, but must use the equivalent of *Mary exited the room running/swimming/dancing/waltzing/parading/skating...*’ or the like (see Levin & Rappaport-Hovav (1995) for a review and further references).

There may also be some relevant limits to the class of attitude verbs that allow propositional OC in French. Two verbs that gave no google hits in my searches were *conjecturer* ‘guess’ and *alleguer* ‘allege’. Judging from English, these are verbs that are particularly unlikely to select CP complements that are about their agents. Indeed, I got no hits for “*J’ai conjecture que je suis/ai*” (‘I conjecture that I am/have...’), just like I got none for “*J’ai conjecture*

avoir/etre” (‘I conjecture to have/be...’). I also got no hits for “*j’ai allégué que je suis/ai*” (‘I allege that I am/have...’), whereas I got (only) three for “*j’ai allégué avoir/etre*” (‘I allege to have/be...’). These verbs then plausibly do not license a “& about(z)=x” term even in French, and as a result do not allow OC. The overall situation can be summarized as in (59). The crosslinguistic difference is in (59c), which may be the largest class of predicates.

(59) a. verbs skewed heavily toward being about the agent (*claim, confess, boast, deny...*):

OC in English, OC in French

b. verbs skewed heavily toward not being about the agent (*conjecture, allege, ...*):

no OC in English or in French

c. Verbs not heavily skewed either way (*say, think, believe, declare...*):

no OC in English, yes OC in French.

It is not clear (to me) how to investigate this hypothesis directly by way of lexical semantics, given that adding the “& about(z)=x” term to the interpretation of a clause headed by ‘believe’, ‘know’ or ‘tell’ is clearly optional. I do not, for example, predict that *croire, penser* and *dire* will resist non-*de-se* interpretations of pronouns in their CP complement, the way that *claim* and *admit* do in English (and rightly so, since non-*de-se* interpretations comparable to (28) are possible in French, Keny Chatain, p.c.). However, the account is testable in that it predicts that OC with all sorts of propositional verbs in French will show the same properties and restrictions that I derived for *claim* in English in section 4.3. This is correct, as I now go on to show.

First, we have seen that *claim* is a subject control predicate, even when it takes an object argument, in defiance of the Minimal Distance Principle. The same is true of the larger class of propositional control verbs in French. (60) shows this for *dire* ‘say’, *affirmer* ‘affirm’ and *révéler* ‘reveal’.

- (60) a. Je lui ai dit avoir fauché un cerf. (Internet)
 ‘I to.him have said to.have (that I have) hit a deer.’
 b. J’ai affirmé au juge avoir fait une grave erreur. (Viviane Déprez, p.c.)
 ‘I-have affirmed to.the judge to.have (that I have) made a serious error.’
 c. Le jour de sa sortie, il m’a révélé avoir une hépatite C... (internet)
 ‘The day of his release, he me-has revealed to.have (that he had) hepatitis C.’

These verbs select for a dative internal argument (a goal), which one might think is syntactically a PP (but probably not). But *informer* ‘inform’ takes an accusative DP object bearing the theme role along with a CP. It also robustly shows subject control even in the presence of an object.

- (61) a. (Marie,) le portier l’a informée avoir trouvé son sac. (4 consultants)
 ‘(As for Marie,) the doorman her-has informed to.have (that he has) found her bag.’
 b. Je l’ai informé être allé chercher une copie du jugement. (internet)
 ‘I him-have informed to.be (that I have) gone to.look.for a copy of.the judgement.’

This seems quite systematic with any communication verb that allows an object in French. I found at least a small number of examples like (60) and (61) with the verbs *expliquer* ‘explain’, *avouer* ‘confess’, *repondre* ‘respond’, *nier* ‘deny’, *spécifier* ‘specify’, *ecrire* ‘write’, *préciser* ‘specify’, and *stipuler* ‘stipulate’. This is not all the verbs that one might imagine doing this, but it is a good selection. So there is a largish class of subject control verbs in this lexical domain in French. This is what I predict, given that OC is triggered by a *res* thematic role shared between the matrix and embedded clause, since this is unaffected by whether there is an additional goal/benefactive (or theme) argument in matrix clause (see (43) for a derivation).²⁵

²⁵ This is another reason not to pursue the hypothesis that OC with propositional verbs in French is induced by there being an epistemic modal in the complement clause. In that case one would expect the subject of the embedded clause to be matched up with the object of the matrix clause if the matrix clause has a causative semantics along the lines of “X caused Y to know that P” (compare *persuade*, meaning “X caused Y to intend to P”, which does have

This analysis also predicts that having a passive in the embedded clause should have no power to shift an example from subject control to object control with propositional verbs; compare the contrast between *promise* and *claim* in English ((44) vs (45)). Indeed, it is not hard to find examples with *informer* or *dire* with a passive complement. All such examples that I have noticed do indeed have subject control rather than object control. (62) gives two examples.

(62) a. Elle lui a dit avoir été frappée par son mari le 18 Fev 2007. (Internet)

‘She to.him has told to.have (that she has) been hit by her husband (on) Feb 18, 2007.’

b. Il m'a informé avoir été contacté par monsieur BOMBA de la DDT. (Internet)

‘He me-has informed to.have been (that he was) contacted by Mr. Bomba of the DDT.’

This also follows from the fact that OC with these verbs depends on aboutness relations rather than on the more familiar thematic roles of agent and goal/beneficiary.

Next, my account predicts that impersonal passives should not be possible for propositional verbs with control complements, in contrast to desiderative verbs like ‘decide’ which do allow impersonal passives. Like its English cognate, *decider* in French allows an impersonal passive:

(63) Il a été décidé de travailler avec beaucoup de bois. (Internet, 4 consultants)

‘It has been decided (of) to.work with a.lot of wood.’

object control, whereas *intend* itself has subject control). *Informer* has the requisite causative semantics, but it still does not have a propositional form of object control.

One consultant does allow an epistemic object control reading a sentence like *Je l'ai informé être un imbécile* (‘I_i him_k-have informed PRO_{i,k} to be an idiot’) alongside the subject control reading, and there are a small smattering of examples in the internet. However, they are outnumbered by subject control examples about 50:1, so it is a more marginal phenomenon. I can include it by saying that the covert *about* phrase associated with a verb like ‘inform’ can marginally be anaphoric to the direct object of the clause rather than to the subject. How possible this really is is a topic for future research.

However, all my consultants reject impersonal passives with propositional verbs, and I found no examples on the internet using the same kind of Google searches.²⁶

(64) a. *Il est souvent pensé avoir un bon sens de l'humour.

'It is often thought to.have a good sense of humor.'

(Meaning: 'People often think that they have a good sense of humor.')

b. *Il est dit n'avoir rien fait de mal.

It is said NEG-to.have anything done of bad

(Meaning: 'People say that they have not done anything wrong.')

c. *Il n'est jamais réalisé être un tyran.

It NEG-is never realized to.be a tyrant

(Meaning: 'One never realizes that one is a tyrant.')

As with English *claim*, this restriction follows from the assumption that the anaphoric aboutness term that OC depends on cannot be bound by a backgrounded expression like an implicit agent.

The last (cluster of) prediction(s) that follows from the *claim*-style analysis of propositional OC in French is that having a distinct explicit representation of the *res* argument associated with the matrix verb should make subject control impossible, as it does for *claim* in English (see (38)). In fact, French has several aboutness constructions, with different levels of generality/productivity (Godard 1988, Koopman & Sportiche 2014; for some discussion of the French possibilities in a broader crosslinguistic context, see Salzmann 2017). However, all of them interfere with propositional OC, as expected. The most straightforward but perhaps the least productive construction is having an overt *de*+DP in postverbal position—a direct analog of

²⁶ Possible cases turn out to be exceptional ECM examples, where *il* 'it, he' is not an expletive pronoun but a referential one, similar to English 'He is believed to be dying' (see Kayne 1984, Rizzi 1982). For example, one consultant marginally accepts (64b) on the meaning 'He is said to have done nothing wrong.' There are also examples of this sort on the internet.

English. This is possible with a relatively small number of verbs, including the most canonical propositional OC verbs *affirmer*, *croire*, *dire*, *penser*, and *savoir* (Godard 1988: 25); (65a) is an example.²⁷ However, where this sort of *de* phrase is possible, it does interfere with subject control, as in (65b).

(65) a. Paul savait de ce dossier qu’il serait difficile à traiter. (Godard 1988: 25)

‘Paul knew of this file that it would.be difficult to handle.’

b. *Paul savait de ce dossier l’avoir déjà traité. (consultants)

‘Paul knows about this file it to.have (that he has) already handled it.’

With other verbs, aboutness can be expressed with a more complex expression such as *au sujet de*, or *apropos de*. These are already awkward with a finite CP complement, perhaps for prosodic reasons. But they are much worse with an infinitival control complement, as predicted.

(66) a. ?Je pense au sujet de/apropos de Guy que je l’ai aidé. (consultants)

‘I think on the subject of Guy that I him-have helped.’

b. ?*Je pense au sujet de/apropos de Guy l’avoir aidé.

‘I think on the subject of Guy him-to.have (that I have) helped.’

For some speakers, the most productive aboutness construction in French seems to be one that is in a relative clause with the genitive complementizer *dont*. This seems to be derived from an underlying form that is like the one seen in (65a), but the ‘about’ (*de*) phrase is reduced to a clitic-like form *dont* and extracted by *wh*-movement (Godard 1988, Koopman and Sportiche 2014). This is possible when the attitude verb takes a finite clause complement, as in (67).²⁸

²⁷ A somewhat wider range of predicates are possible with a *de* ‘about’ phrase as long as that phrase is fronted to initial position in the sentence (Godard 1988: 27-28; Koopman and Sportiche 2014). I haven’t investigated this variant, but predict that a fronted *de* phrase will also interact negatively with OC.

²⁸ One of my consultants (a linguist) does not really like this construction at all, except perhaps as a last resort. Another says that it has the flavor of being both “new” and “literary”. All found (67b) better than Godard’s (67a). But crucially all roundly rejected all the examples in (68)—which I found on the internet, but which seems to be quite rare there.

(67) a. un incident dont le directeur suggère que ses subordonnés l'ignorent (Godard 1988)
 'an incident about-which the director suggests that his subordinates ignore it'

b. un professeur dont les parents savent que les élèves lui font confiance.
 'a professor about-which the parents know that the students in.him have trust'

However, everyone finds this gapless *dont* construction very bad with a nonfinite clause complement with OC:

(68) a. *ton superbe livre dont je crois l'avoir déjà vu quelque part (4 consultants)
 'your superb book about-which I believe it-to.have seen some part'

b. *ce parfum dont j'ai dit l'avoir inhalé depuis un an et demi
 'this perfume about-which I-had said it-to.have inhaled for one year and (a) half'

c. *une rayure dont je doute l'avoir fait
 'a scratch [in a rental car] about-which I doubt it-to.have made'

Indeed, Godard (1988: 65) notices this effect (although she does not emphasize the fact that the nonfinite version has OC), citing contrasts like the following:

(69) a. un texte dont tu es certain que tu le résoudras rapidement (finite CP, coreference)
 'a text about-which you are certain that you it-will.solve quickly'

b. *un texte dont tu es certain de le comprendre (infinitive CP, control)
 'a text about-which you are certain of it-to.understand (i.e., ...that you understand it)'

This also supports the *claim*-like analysis: whenever there is an overt (nonanaphoric) realization of the *res* argument, it is impossible for the matrix subject to control PRO inside the infinitival complement. That is because a unique covert anaphoric *res* argument must mediate this control.

As a coda to this point, I mention that Godard (1988) shows that the gapless *dont* construction is possible with a wide range of predicates in French (see also Koopman & Sportiche 2014: 62). (70) gives two additional examples.

- (70) a. une difficulté dont Paul est certain qu'il le résoudra
 'a difficulty about.which Paul is sure that he it-will.solve'
 b. un texte dont il semblait que certains élèves ne l'avaient même pas lu
 'a text about.which it seems that some students NEG it-had even not read'

In contrast, these predicates do not allow an *about* phrase in (my) English, so (71a,b) are bad:

- (71) a. Paul is sure (??about that difficulty) that he will be able to solve it.
 b. It seems (*about that text) that some students have not even read it.

However, for purposes of propositional OC in French, it does not matter that the English predicates do not allow a *res* argument; it only matters that the French ones evidently do, in this language-specific form. Therefore, I predict that these predicates will also allow OC in French. And in fact, they do, as seen in (72).

- (72) a. Je suis certain avoir vu une annonce sur les 2 groupes Roxton. (internet)
 'I am sure to.have (that I have) seen an ad about the two Roxton groups.'
 b. Il me semble avoir compris. (Kayne 1984: 119 n.7)
 'It to.me seems to.have (that I have) understood.'

This supports the notion that there is a relationship between the particular aboutness constructions that are compatible with particular propositional predicates in a given language and the possibility of propositional OC with those predicates in that language—as my analysis would expect. It would be worthwhile to follow up by studying in finer detail which verbs allow the gapless *dont* construction and which do not in French, but I leave this to future work (probably by someone more knowledgeable about French than me).²⁹

²⁹ It may be too strong to predict that any French attitude verb that allows the gapless *dont* construction will also allow OC with an infinitival complement. There may be verbs that allow an '& about(e)=x' term with a given predicate but which resist the *about* term being anaphoric to the subject of the clause. See the discussion of 'conjecture' and 'allege' above.

6. Conclusion

This paper has considered the question of why certain verbs cannot appear with obligatory control complements—even when they select nonfinite clauses that have suitable tense-aspect properties. It has done so by focusing on propositional verbs like *say* and *believe* in English, which do not allow OC. Its most fundamental claim is that OC must be licensed by thematic uniqueness; it arises when and only when a DP in the matrix clause and PRO in the complement clause bear the same thematic role to the same eventuality, following Baker (2024). This condition is satisfied by many desiderative verbs in English (among others), but not by most propositional verbs in English.

This paper has also considered why some propositional verbs do in fact allow OC, including *claim* in English and a large variety of propositional verbs in French. The proposal is that these verbs allow a covert *res* (‘about’) argument that is anaphoric to the verb’s subject argument. This then overlaps with the fact that the embedded clause can be understood as being about its PRO subject, such that the two expressions must have the same semantic value by (a generalized version of) Carlsonian thematic uniqueness. This account is grounded in certain lexical semantic judgments and explains a wide range of ways in which control with propositional verbs is different from control with more familiar desiderative verbs like *decide*: the fact that propositional verbs favor subject control even in the presence of an object, the fact that passive in the complement of a propositional verb never induces control shift, the fact that an overt ‘about’ phrase blocks OC with a propositional verb, and the fact that propositional control structures do not allow impersonal passives. In these grammatical respects, French and English are remarkably similar, despite the fact that the propositional OC construction is found

with many more verbs in French than in English. This points to the conclusion that the fundamental grammatical factors that govern OC are stable across languages, whereas the lexical semantic associations of similar verbs can vary systematically within a certain range. This supports a theory in which obligatory control relationships depend on thematic uniqueness.

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